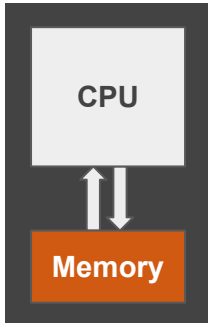
Variables, DataTypes, Arithmetic Operators and Expression

اَللّٰهُمَّ اُرْزُقْنِی عِلْمًا نَافِعًا وَاسِعًا عَمِيْقًا

اَللّٰهُمَّ اُرْزُقْنِى رِزْقًا وَاسِعًا حَلَالًا طَيِّبًا
مُبَارَكًا مِنْ عِنْدِكَ

Memory and Variables

Review: Memory

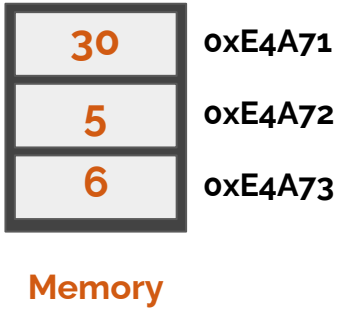


- When CPU takes input from devices, it stores information into **memory** before processing it.
- CPU stores results of the processing into the **memory.**
- CPU stores information into the **memory** before sending it to output devices.

How to Allocate Memory: Variables

To store data into the Memory, we need to reserve the space in the Memory.

When the space is reserved, we can store or retrieve data from the Memory through its Memory Addresses.

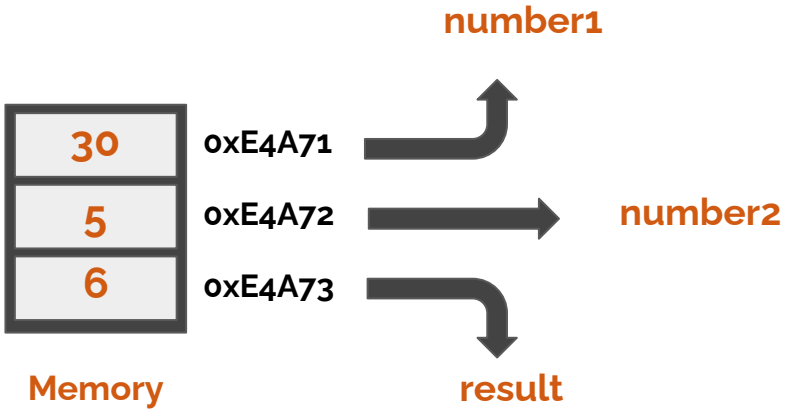


Variable Name

Variables: Names

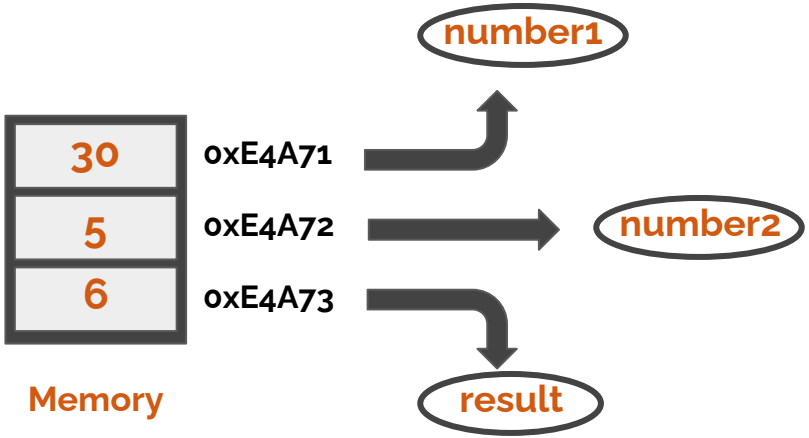
It is difficult to remember the Addresses of these Memory locations.

High Level Languages allow us to give Names to these reserved Memory locations.



Variables: Names

These Names are called the Variables.
Variables are the names that we give to the Memory Locations.



Variables: Names

All High Level Languages apply some Naming Rules on the variables

- The name can not have Spaces
- The name can not start with Numbers
- The name can not have any Special Character (&, !, %, # etc)

Variables: Names

All High Level Languages apply some Naming Rules on the variables

These are some of the Valid names of the Variables

number1 num_1 num1

numb2 nu_2 _n2

result_1 Res _Res numOne

Variable Datatypes

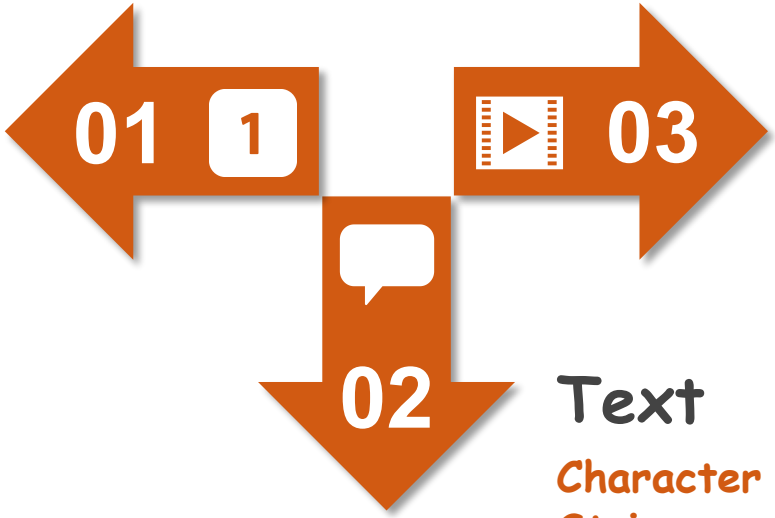
What Type of Data in Memory?

Now, We know that we can deal with memory using Variables. But the question is What type of Data is in memory?

Variables: Types of Data

Numbers

Integer
Float
Double
Boolean



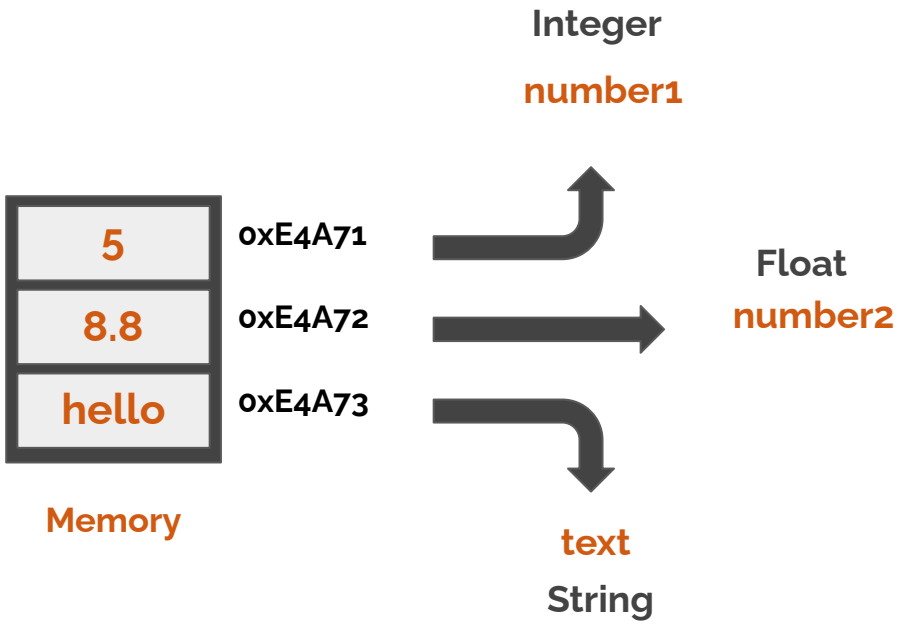
Text

Character
String

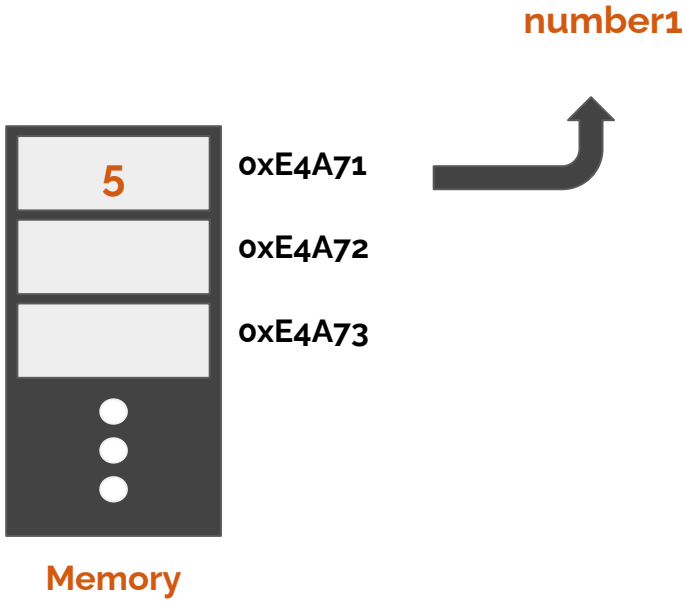
Multimedia

Image
Audio
Video

Variables: What Kind of Data Inside



Variables: What Kind of Data Inside



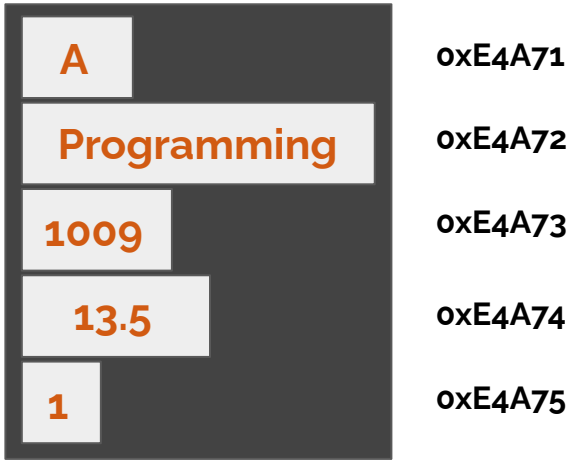
Data Types: Why Inform Memory

Data Types: Why Inform Memory

- To Adjust Size of Allocated Memory Cell
- To Check the Validity of the Operations

Data Types: Size of Memory

Different types of data require Different sizes of cells in memory.



Memory

Data Types: Validity of Operations

We also need to Check whether an Operation applied on the data is Valid or Not.

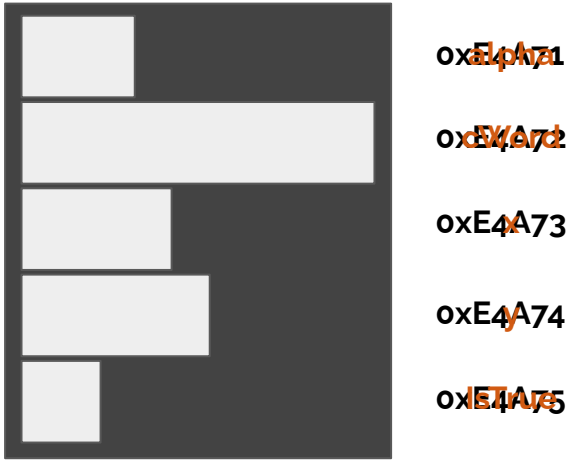
For Example:

- 20 + 20.5

- 20 + Programming

Variable Declaration: Reserve Memory

Reserving the memory location through Variables for certain type of data is also called Variable Declaration.

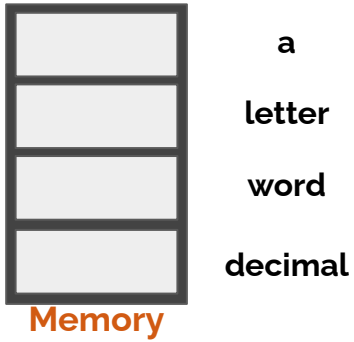


Memory

Variable Declaration: Reserve Memory

In many High Level Languages like C++, Java and C# the variable declaration is done as

```
Datatype nameOfTheVariable;
int a;
char letter;
string word;
float decimal;
```



Uses of Variables

- assign values to these variables according to their
- data types
- retrieve values from these variables
- We can apply different mathematical addition,multiplication,subtraction) and other operations (we will see those in next lecture) on these variables.

Uses of Variables: Assignment

We can Assign a value to variable using Assignment Operator.

01

Constants

```
int x = 8;
```

02

Variables

```
int y = x;
```

03

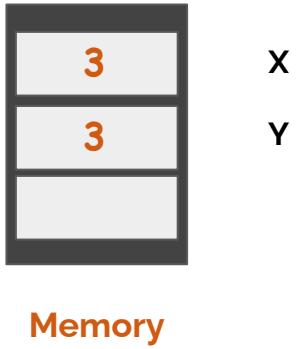
Expressions

```
int z = 5 * 2 / x;
```

Uses of Variables: Retrieval

Here, we are Retrieving the value of variable x and assigning that value to variable y

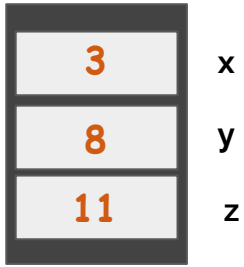
```
int X = 3;
int Y = x;
```



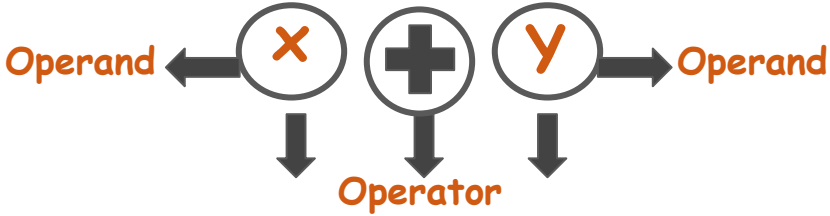
Operations on Variables: Addition

Apply mathematical operation on Variables

```
int x = 3;
int y = 8;
int z = x + y;
```



Memory



11 ← 3 + 8

Arithmetic Operators:

Here is a list of **Arithmetic Operators** that can be used.

Operator	Meaning	Example
+	Addition	8+2=10
-	Subtraction	8-2=6
*	Multiplication	8*2=16
/	Division	8/2=4
%	Modulus	8%2=0

Arithmetic Operators:

These operators also called **Binary Operators**.
Why ?

Operator	Meaning	Example
+	Addition	8+2=10
-	Subtraction	8-2=6
*	Multiplication	8*2=16
/	Division	8/2=4
%	Modulus	8%2=0

Expressions

An **Expression** is a combination of **Variables**, **Constants** and **Operators**.

For Example

- **8 + 9** is an expression
- **X/2 – 1** is also an expression

Expressions

It consists of

- One or more Operands
- Zero or more Operators

For Example:

X + 10 - Y

Operands

Operations on Variables: Expression

Expression containing only Constants and Operators

$$2 + 10 + 8$$

Operations on Variables: Expression

Expression containing combination of Variables, Constants and Operators

27 ← X + 10 + Y

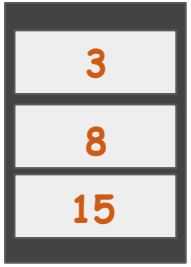
X → 5

Y → 12

Expressions

We can write Expression using Variables and Constants and Assign these Expressions to some Variables.

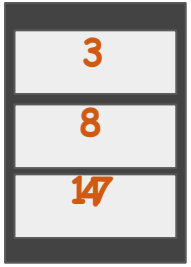
```
int x = 3;
int y = 8;
int z = y-x+10;
```



Expressions

Lets see some more examples

```
int x = 3;
int y = 8;
int z = 2y/25 + 7;
```



x

y

z

Memory

Operations on Variables: Expression

$$Z = 10 + 10 * 5$$

Operations on Variables: Expression

$$60 = 10 + 50$$

$$Z = 10 + 10 * 5$$

$$100 = 20 * 5$$

Operations on Variables: Expression

Z = 60

Z = 10 + 10 * 5

~~Z = 100~~

Operations on Variables: Expression

$$Z = 60$$

$$Z = (10 + 10) * 5$$

$$Z = 100 * 5$$

Expression : Precedence Order

Here is the precedence order of Arithmetic Operators

Operator	Symbol	Precedence
Parentheses	( )	1
Exponential	X^y	2
Multiplication Division	* /	3 3
Addition Subtraction	+ -	4 4

Expression: PEMDAS RULE

Simply, we can Remember the order of precedence through the PEMDAS Rule.

Rule	Operator	Symbol
P	Parentheses	( )
E	Exponential	X^y
M D	Multiplication Division	* /
A S	Addition Subtraction	+ -

Working Examples: Expressions

Lets see some working examples of Expressions

$$Z = 2 + 3 / 4$$

$$Z = 2.75 \quad 0.75$$

Working Examples: Expressions

Lets see some working examples of Expressions

$$Z = 10 - 2 * 4$$

$$Z = 2\,10 - 8$$

Working Examples: Expressions

Lets see some working examples of Expressions

$$Z = 4 + 100 / 20 * 2$$

Working Examples: Expressions

How to solve?

$$Z = 4 + 100 / 20 * 2$$

Working Examples: Expressions

When there are operators with same precedence order then we solve the expression from left to right (Associativity)

$$Z = 4 + 100 / 20 * 2$$

Working Examples: Expressions

When there are operators with same precedence order then we solve the expression from left to right (Associativity)

$$Z = 4 + 100 / 20 * 2$$

Working Examples: Expressions

When there are operators with same precedence order then we solve the expression from left to right (Associativity)

$$Z = 4 + 100 / 20 * 2$$

$$\downarrow$$

$$Z = 4 + 5 * 2$$

Working Examples: Expressions

When there are operators with same precedence order then we solve the expression from left to right (Associativity)

$$Z = 4 + 100 / 20 * 2$$

$$\downarrow$$

$$Z = 4 + 5 * 2$$

$$\downarrow$$

$$Z = 4 + 10$$

Working Examples: Expressions

When there are operators with same precedence order then we solve the expression from left to right (Associativity)

$$Z = 4 + 100 / 20 * 2$$

$$\Downarrow$$

$$Z = 14$$

Learning Objective

Write expression using **Arithmetic Operator**, **Variables**, and **constants** while following precedence rule.

Self Assessment

1. What is a Variable?
2. How we can store and load data from the Memory using variables?
3. From the given table below, tell which Variable Names are Valid and which are not.

Variable	Valid/Invalid
mul*	
Foo	
Do it	

Self Assessment

4. Define Variable Declaration. And Declare a variable to store a value of 58.9

5. Write the Datatypes of the following data given in the table

Data	Datatype
400.6	
My name is Kaka	
C	
12	

6. Declare the variables to store the above mentioned data in the variables.

Hint: float a; (a is a variable that will store float type of data)

Self Assessment

1. Find constant, variable and operator from the following statements

Statement	Constant	Variable	Operator
Foo = 4 * result			
Var = 5 % 3			
X = num1 - num2			

2. Solve the following Expressions and write the answer.

Statement	Answer
Foo = 4 * 10 / 2	
Var = 5 % 3	
X = 5 - 2 + 62 - 2	

Self Assessment

3. Evaluate the following expressions and Write the answers.

No.	Expression	Answer
1	2 / 1 + 5	
2	3 / 4 + (2 - 1)	
3	7 + (600 - 100) * 8	
4	500 * 400 / 4 + 10	
5	18 / 2 * 18 - 1	

Conclusion

- We can have multiple uses of variables
  1. Assign Values 2. Retrieve Values 3. Apply Mathematical Operations
- Assignment is done using Assignment Operator.
- There are 3 ways in which we can assign values to the variables
  1. Constants 2. Variables 3. Expressions
- An Expression is a combination of Variables, Constants and Operators.
- Expressions are evaluated with the Precedence order of Operators.
- The precedence order is given by PEMDAS Rule.
- If operators are with same precedence order then they are solved from left to right (Associativity).